Data-Backed Journal-Style Replication of *Beyond Local Search*

Replication Paper Project

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Abstract

This paper presents a local calculated replication of Beyond Local Search: Boundary-Spanning, Exploration, and Impact in the Optical Disk Industry using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The current local batch contains 23,844 observations across 184 entities, and the batch headline estimate is -0.0389 for boundary_existing_field vs local with p = 0.3868. The local analogue supports the most novel boundary-spanning channel much more clearly than the moderate boundary-spanning channel. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: boundary-spanning search; exploration; inventive impact; patent classes; local calculated analogue

1 Introduction

This replication begins from a paper-first reading of Beyond Local Search: Boundary-Spanning, Exploration, and Impact in the Optical Disk Industry and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The motivating question is: The source paper asks whether boundary-spanning search improves inventive impact relative to local search. The study is worth revisiting because The replication matters because boundary-spanning search claims are often treated broadly, while the original paper distinguishes among qualitatively different

forms of exploratory search. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted. The executable evidence base used in this manuscript contains 23,844 observations across 184 entities and spans 1986 to 2016 in the local executable panel.. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets. The paper's contribution is a transparent three-way search taxonomy on a local executable panel that separates local, boundary-existing-field, and boundary-new-field search. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions.

2 Original Study, Theory, and Replication Boundary

The focal source article is [1]. The original study can be summarized as a mechanism paper with a sharp empirical core. The source paper contrasts local search with boundary-spanning exploration. Impact rather than patenting volume is the main empirical outcome. Industry-specific coding is part of the original design. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data. In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue. The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design.

2.1 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The source paper emphasizes different kinds of boundary-spanning search. The local reconstruction therefore adopts the following bounded decision: The local panel reconstructs a three-way search taxonomy from experience counts. This bridge is treated as a visible design choice, not as a hidden substitution. The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Impact is the key outcome. The local reconstruction therefore adopts the following bounded decision: The outcome remains impact-focused through citations and a top-tail indicator. This bridge is treated as a visible design choice, not as a hidden substitution. The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Industry-specific coding is part of the original design. The local reconstruction therefore adopts the following bounded decision: The industry-specific coding is generalized to the processed multi-firm patent panel. This bridge is treated as a visible design choice, not as a hidden substitution.

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Patent-level recombination panel built from the cleaned reproduction dataset. Search type reconstructed from prior organizational and field experience counts. Impact outcomes measured with patent-level inventive impact and a top-5-percent indicator. Operationally, the reconstruction moves through the following file chain: Analysis panel is backed by patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_analysis_dataset.csv; Coefficient export is backed by patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_model_coefficients.csv; Sample summary is backed by patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_sample_summary.csv. The resulting executable panel supplies

23,844 observations across 184 entities and covers 1986 to 2016 in the local executable panel.. Every numeric claim in the results section is intended to trace back to one of those verified files rather than to a hand-copied number from the article. This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated. The local analogue reconstructs local, boundary-existing-field, and boundary-new-field search from organizational and field experience counts already materialized in the processed recombination panel. The local sample frame is Patent-level recombination panel derived from the cleaned local reproduction dataset. The observed time window in the executable panel is 1986 to 2016 in the local executable panel. The outcome is Patent-level inventive impact and a yearly top-tail impact indicator. The main executable panel is stored at patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_analysis_dataset.csv and the principal coefficient export used in the paper is stored at patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_model_coefficients.csv . The search-type file classifies patents into local, boundary-existing-field, and boundary-new-field search. Local patents account for 12,131 observations with mean impact 5.1357. Boundary-existing-field patents account for 3,270 observations with mean impact 4.5352. Boundary-new-field patents account for 8,443 observations with mean impact 5.8602. Even before estimation, the raw means point to a substantive ranking in which the most novel form of search outperforms the more moderate boundary-spanning category. The top-tail shares follow the same direction: 0.0411 for local search, 0.0287 for boundary-existing-field search, and 0.0458 for boundary-new-field search.

3.1 Variable Construction and Derivation

The construct Local search is operationalized locally as Prior organizational experience and appears in the executable dataset as search_type = local. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Classify patents with positive prior organizational experience as local.. This mapping is retained because it preserves the construct's functional

role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The construct Boundary-existing-field search is operationalized locally as No prior organizational but prior field experience and appears in the executable dataset as search_type = boundary_existing_field. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Classify patents with zero prior organizational but positive field experience as boundary-existing-field.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The construct Boundary-new-field search is operationalized locally as No prior organizational and no prior field experience and appears in the executable dataset as search_type = boundary_new_field. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Classify patents with zero prior organizational and zero prior field experience as boundary-new-field.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The construct Inventive impact is operationalized locally as Patent-level citation impact and appears in the executable dataset as inventive_impact. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the local patent-level impact variable from the cleaned panel.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The construct Tail impact is operationalized locally as Top-5-percent yearly impact indicator and appears in the executable dataset as high_top5_y. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the already materialized yearly top-5-percent indicator.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation.

4 Empirical Strategy and Identification

The estimand targeted in this manuscript is Patent-level impact and top-tail models with search-type indicators and patent-level controls. The role of the empirical strategy is therefore to recover

the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism. Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest. The first set of replication decisions concerns scope. The unit of analysis is Patent, the sample frame is Patent-level recombination panel derived from the cleaned local reproduction dataset., and the empirical window is 1986 to 2016 in the local executable panel.. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample. The second set of decisions concerns measurement and specification. The outcome is Patent-level inventive impact and a yearly top-tail impact indicator., and the main explanatory design is Patent-level impact models with search-type indicators comparing boundary-spanning forms to local search.. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question. The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Patent-level controls for breadth, degree, citations, scale, and prior stock with available fixed effects.. The crucial exclusion or substitution rule is: The original industry-specific coding is not locally recoverable, so the executable rerun generalizes the taxonomy using prior organizational and field experience counts.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability. A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. When the local data architecture differs, the paper marks the result as a calculated analogue rather than claiming byte-identical duplication.

$$E[\text{Impact}_{pft} \mid X_{pft}] = \exp\left(\beta_1 \nVdash[\text{BoundaryExisting}_p] + \beta_2 \nVdash[\text{BoundaryNew}_p] + \beta_3 X_{pft} + \alpha_f + \gamma_t\right). \tag{1}$$

5 Results

5.1 Main Estimates

The central local estimate is -0.0389 on boundary_existing_field vs local (p = 0.3868). The local analogue supports the most novel boundary-spanning channel much more clearly than the moderate boundary-spanning channel. Boundary new field versus local is estimated at 0.1126 with p = 0.0014. Recombination degree is estimated at -0.4108 with p = 0.0226. Prior-art citations is estimated at 0.0013 with p = 0.0548. Table 1 reports the compact coefficient pattern most relevant to the source paper's central claim. The broader coefficient dump, descriptive statistics, grouped summaries, and audit material are retained in the appendix so that the main text can focus on interpretation.

model	Term	estimate	std_error	p_value
rosenkopf_impact	Boundary existing field versus local	-0.0389	0.0449	0.3868
rosenkopf_impact	Boundary new field versus local	0.1126	0.0353	0.0014
rosenkopf_impact	Breadth of recombination	0.0223	0.0130	0.0870
rosenkopf_impact	Recombination degree	-0.4108	0.1802	0.0226
rosenkopf_impact	Prior-art citations	0.0013	0.0007	0.0548
rosenkopf_impact	Log prior patent stock	-0.1965	0.1067	0.0657

Key coefficient table used in the main text.

5.2 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Boundary new field versus local is estimated at 0.1126 with a p-value of 0.0014; Recombination degree is estimated at -0.4108

with a p-value of 0.0226; Prior-art citations is estimated at 0.0013 with a p-value of 0.0548; Log prior patent stock is estimated at -0.1965 with a p-value of 0.0657 . Within the focal specification, the sharpest local signal comes from Boundary new field versus local, which is estimated at 0.1126 with p = 0.0014. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment. The scale of the underlying variables also matters for interpretation. The outcome Inventive impact has mean 5.3099 and standard deviation 17.5532 in the local panel, while the main regressor Breadth of recombination has mean 4.2941 and standard deviation 4.5669. A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 4.4643 to 8.1151 while the mean regressor ranges from 2.4112 to 9.9635. This does not substitute for the model, but it shows how much raw variation the specification must explain. Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local analogue supports the most novel boundary-spanning channel much more clearly than the moderate boundary-spanning channel. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation.

6 Comparison to the Published Study

The comparison to the published paper is intentionally narrow. The local evidence is strongest for boundary-new-field search rather than for boundary-existing-field search. Boundary-new-field patents receive higher average impact relative to local search, while boundary-existing-field patents are lower and imprecise. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication. This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation.

7 Discussion

The local evidence is strongest for boundary-new-field search rather than for boundary-existing-field search. Boundary-new-field patents receive higher average impact relative to local search, while boundary-existing-field patents are lower and imprecise. Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly. The first limitation concerns measurement. The local search-type coding generalizes the original industry-specific implementation. The second limitation concerns sample translation. Mismatch with published estimates can arise from both sample expansion and measurement differences. The third limitation concerns interpretation. The exercise is most informative about directional mechanisms.

8 Conclusion

The paper's contribution is a transparent three-way search taxonomy on a local executable panel that separates local, boundary-existing-field, and boundary-new-field search. The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available.

A Appendix A: Reading Protocol

This appendix records the paper-first workflow followed before any manuscript text was written. The sequence starts from the source article, then maps required empirical objects to verified local files, and only then proceeds to estimation and interpretation.

Step	Action	Evidence
1	Read the original paper before touching the local estimates.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/patent literature/Strategic Management Journal - 2001 - Rosenkopf - Beyond local search boundary-spanning exploration and impact in the.pdf
2	Extract the source paper's theoretical logic, sample definition, variable definitions, and estimand.	Beyond Local Search: Boundary-Spanning, Exploration, and Impact in the Optical Disk Industry

Step	Action	Evidence
3	Match each required empirical object to a local executable file and verify that the file exists.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_analysis_dataset.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_model_coefficients.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_sample_summary.csv
4	Write the replication paper only from the checked local files and calculated outputs.	The manuscript, markdown notes, and PDF are generated from the verified file set.

Paper-first protocol followed before writing the manuscript.

B Appendix B: Data Lineage and File Manifest

This appendix documents the data lineage as a sequence rather than as a pile of filenames. The source article anchors the theory and published design. The executable local files then provide the panel, coefficient export, sample reconstruction log, and, where available, fit statistics used in the manuscript. Source article: patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/patent literature/Strategic Management Journal - 2001 - Rosenkopf - Beyond local search boundary-spanning exploration and impact in the.pdf. Its role in the manuscript is Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices. Analysis panel: patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_analysis_dataset.csv. Its role in the manuscript is Estimate the local model and regenerate descriptive statistics. Coefficient export: patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_model_coefficients.csv. Its role in the manuscript is Recover the reported local coefficients and p-values. Sample summary: patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_sample_summary.csv. Its role in the manuscript is Audit sample construction and raw grouped moments. The purpose of laying out this chain in prose is to show that the manuscript's claims travel through a verified sequence of evidence. In other words, the paper does not jump directly from the published theory to the final coefficient table. It moves through the exact local files needed to rebuild the analysis for Rosenkopf and Nerkar (2001).

File	Role
rosenkopf_2001_analysis_dataset.csv	Patent-level search-type panel
rosenkopf_2001_model_coefficients.csv	Impact and top-5 coefficient table
rosenkopf_2001_sample_summary.csv	Search-type raw means

Compact list of core local files used for this manuscript.

Stage	NeededFor	FilePath	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/patent literature/Strategic Management Journal - 2001 - Rosenkopf - Beyond local search boundary-spanning exploration and impact in the.pdf	No	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_analysis_dataset.csv	Yes	Executable local dataset

Stage	NeededFor	FilePath	Exists	Status
Coefficient export	Recover the reported local coefficients and p-values	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_model_coefficients.csv	Yes	Executable local result file
Sample summary	Audit sample construction and raw grouped moments	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_sample_summary.csv	Yes	Executable local result file

Source and local file manifest for the manuscript.

C Appendix C: Replication Boundary and Decision Register

Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Patent inside the sample frame Patent-level recombination panel derived from the cleaned local reproduction dataset. over 1986 to 2016 in the local executable panel.. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Patent-level inventive impact and a yearly top-tail impact indicator. and the main explanatory design is Patent-level impact models with search-type indicators comparing boundary-spanning forms to local search.. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Patent-level controls for breadth, degree, citations, scale, and prior stock with available fixed effects. The main exclusion or substitution rule is The original industry-specific coding is not locally recoverable, so the executable rerun generalizes the taxonomy using prior organizational and field experience counts. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

DirectSourceFact	LocalReplicationDecision
The source paper emphasizes different kinds of boundary-spanning search.	The local panel reconstructs a three-way search taxonomy from experience counts.
Impact is the key outcome.	The outcome remains impact-focused through citations and a top-tail indicator.

DirectSourceFact	LocalReplicationDecision
Industry-specific coding is part of the original design.	The industry-specific coding is generalized to the processed multi-firm patent panel.

Direct source facts versus explicit local replication decisions.

DecisionPoint	LocalSpecification
Research question	The source paper asks whether boundary-spanning search improves inventive impact relative to local search.
Unit of analysis	Patent
Sample frame	Patent-level recombination panel derived from the cleaned local reproduction dataset.
Time window	1986 to 2016 in the local executable panel.
Outcome definition	Patent-level inventive impact and a yearly top-tail impact indicator.
Main explanatory design	Patent-level impact models with search-type indicators comparing boundary-spanning forms to local search.

DecisionPoint	LocalSpecification
Controls and fixed effects	Patent-level controls for breadth, degree, citations, scale, and prior stock with available fixed effects.
Exclusion or substitution logic	The original industry-specific coding is not locally recoverable, so the executable rerun generalizes the taxonomy using prior organizational and field experience counts.

Expanded decision register for the manuscript.

D Appendix D: Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

SearchType	Patents	MeanImpact	Top5Share
boundary_new_field	8443.0000	5.8602	0.0458
local	12131.0000	5.1357	0.0411
boundary_existing_field	3270.0000	4.5352	0.0287

Detailed sample reconstruction table.

E Appendix E: Variable Mapping and Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Rosenkopf and Nerkar (2001). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Local search, the manuscript uses Prior organizational experience under the exact local implementation search_type = local. The derivation step is: Classify patents with positive prior organizational experience as local.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Boundary-existing-field search, the manuscript uses No prior organizational but prior field experience under the exact local implementation search_type = boundary_existing_field. The derivation step is: Classify patents with zero prior organizational but positive field experience as boundary-existing-field.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Boundary-new-field search, the manuscript uses No prior organizational and no prior field experience under the exact local implementation search_type = boundary_new_field. The derivation step is: Classify patents with zero prior organizational and zero prior field experience as boundary-new-field.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Inventive impact, the manuscript uses Patent-level citation impact under the exact local implementation inventive_impact. The derivation step is: Use the local patent-level impact variable from the cleaned panel.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Tail impact, the manuscript uses Top-5-percent yearly impact indicator under the exact local implementation high_top5_y. The derivation step is: Use the already materialized yearly top-5-percent indicator.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Local search	Prior organizational experience	search_type = local	Classify patents with positive prior organizational experience as local.
Boundary-existing-field search	No prior organizational but prior field experience	search_type = boundary_existing_field	Classify patents with zero prior organizational but positive field experience as boundary-existing-field.
Boundary-new-field search	No prior organizational and no prior field experience	search_type = boundary_new_field	Classify patents with zero prior organizational and zero prior field experience as boundary-new-field.
Inventive impact	Patent-level citation impact	inventive_impact	Use the local patent-level impact variable from the cleaned panel.

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Tail impact	Top-5-percent yearly impact indicator	high_top5_y	Use the already materialized yearly top-5-percent indicator.

Detailed construct mapping.

F Appendix F: Descriptive Statistics

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	M
Inventive impact	23844.0000	5.3099	17.5532	1.0000	2.0000	6.0000	0.0000	812.00
Top-5 indicator	23844.0000	0.0411	0.1985	0.0000	0.0000	0.0000	0.0000	1.00
Breadth of recombination	23844.0000	4.2941	4.5669	2.0000	3.0000	5.0000	2.0000	117.00
Recombination degree	23844.0000	0.5549	0.1738	0.4444	0.5600	0.6914	0.0454	0.95
Field experience	23844.0000	1.6339	1.8203	0.0000	1.0986	2.7726	0.0000	8.01
Organizational experience	23844.0000	0.9227	1.3174	0.0000	0.6931	1.3863	0.0000	7.65
Prior-art citations	23844.0000	15.0909	50.4465	5.0000	8.0000	13.0000	2.0000	2263.00
Same-year patents	23844.0000	728.7229	1302.8291	109.0000	313.0000	652.0000	1.0000	7321.00
Prior cumulative stock	23844.0000	7.5074	2.2142	6.3297	7.6930	8.8934	0.0000	11.32

Descriptive statistics appendix.

G Appendix G: Correlation Matrix

Variable	Inventive impact	Top-5 indicator	Breadth of recombination	Recombination degree	Field experience	Organizational experience	Prior-art citations	Same-year patents	Prior cumulative stock
Inventive impact	1.000	0.455	0.364	0.018	-0.021	-0.008	0.591	-0.018	0.022
Top-5 indicator	0.455	1.000	0.129	0.024	-0.012	0.006	0.149	-0.015	0.005
Breadth of recombination	0.364	0.129	1.000	0.466	-0.263	-0.144	0.822	0.011	0.149
Recombination degree	0.018	0.024	0.466	1.000	-0.456	-0.313	0.140	0.065	0.122
Field experience	-0.021	-0.012	-0.263	-0.456	1.000	0.838	-0.077	0.060	0.132
Organizational experience	-0.008	0.006	-0.144	-0.313	0.838	1.000	-0.018	0.152	0.267
Prior-art citations	0.591	0.149	0.822	0.140	-0.077	-0.018	1.000	-0.016	0.081
Same-year patents	-0.018	-0.015	0.011	0.065	0.060	0.152	-0.016	1.000	0.520
Prior cumulative stock	0.022	0.005	0.149	0.122	0.132	0.267	0.081	0.520	1.000

Correlation matrix for the main numeric variables.

H Appendix H: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	V1	V2	V3	V4	V5
1986.0000	643.0000	4.0202	2.8694	0.5175	0.4277
1987.0000	789.0000	4.2801	2.9176	0.5177	0.4335
1988.0000	1066.0000	4.4606	3.1323	0.5319	0.4353
1989.0000	1390.0000	3.9338	3.0813	0.5272	0.4007
1990.0000	1349.0000	4.5137	3.1156	0.5218	0.3744
1991.0000	1325.0000	4.7381	3.1502	0.5314	0.3743
1992.0000	1302.0000	5.0154	3.3318	0.5346	0.4040
1993.0000	1354.0000	5.6721	3.4609	0.5323	0.3877
1994.0000	1330.0000	6.2654	3.5782	0.5413	0.3932
1995.0000	1217.0000	6.2597	3.8439	0.5563	0.3780
1996.0000	659.0000	6.7269	3.8376	0.5306	0.3763
1997.0000	646.0000	6.0139	4.2446	0.5521	0.4195
1998.0000	356.0000	6.2949	3.9242	0.5473	0.4045
1999.0000	654.0000	5.4832	3.8104	0.5436	0.3700
2000.0000	636.0000	5.5409	4.0220	0.5402	0.3632
2001.0000	614.0000	5.1515	4.4137	0.5625	0.4169
2002.0000	632.0000	4.8892	4.3845	0.5578	0.3687
2003.0000	522.0000	4.5019	4.4176	0.5505	0.3487
2004.0000	324.0000	3.6636	4.8086	0.5734	0.4136
2005.0000	284.0000	4.5000	5.4859	0.5763	0.4085
2006.0000	336.0000	4.8899	5.2649	0.5661	0.3214
2007.0000	246.0000	3.6545	5.2683	0.5692	0.3089
2008.0000	535.0000	3.6523	4.5121	0.5904	0.3607
2009.0000	206.0000	4.8010	6.0049	0.5992	0.2961

Period	V1	V2	V3	V4	V5
2010.0000	511.0000	4.4168	6.6477	0.6078	0.2564
2011.0000	651.0000	6.4516	6.2750	0.6090	0.2550
2012.0000	1068.0000	4.0000	6.5187	0.6113	0.2603
2013.0000	868.0000	5.3859	6.7304	0.5978	0.2604
2014.0000	725.0000	12.6428	6.9890	0.6000	0.2290
2015.0000	1072.0000	6.6157	6.1744	0.5971	0.2043
2016.0000	534.0000	3.7678	4.6292	0.5811	0.1667

Year-by-year sample audit.

I Appendix I: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
1.0000	13675.0000	2.4112	4.4643
2.0000	3458.0000	4.0000	4.9251
3.0000	2273.0000	5.0000	5.5055
4.0000	4438.0000	9.9635	8.1151

Quantile summary for the main regressor.

SearchType	Patents	MeanImpact	Top5Share
boundary_new_field	8443.0000	5.8602	0.0458
local	12131.0000	5.1357	0.0411
boundary_existing_field	3270.0000	4.5352	0.0287

Raw group means from the local summary file.

J Appendix J: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	statistic	p_value	model
search_type::boundary_existing_field	-0.0389	0.0449	-0.8655	0.3868	rosenkopf_impact
search_type::boundary_new_field	0.1261	0.0353	3.1882	0.0014	rosenkopf_impact
breadth_recombination	0.0223	0.0130	1.7114	0.0870	rosenkopf_impact
recombination_degree	-0.4108	0.1802	-2.2800	0.0226	rosenkopf_impact
n_prior_art_cites	0.0013	0.0007	1.9206	0.0548	rosenkopf_impact
total_patents_same_year	-0.0001	0.0000	-2.0994	0.0358	rosenkopf_impact
total_patents_until_prev_log	-0.1965	0.1067	-1.8408	0.0657	rosenkopf_impact
search_type::boundary_existing_field	-0.2317	0.1623	-1.4276	0.1534	rosenkopf_top5
search_type::boundary_new_field	0.1039	0.1131	0.9183	0.3585	rosenkopf_top5
breadth_recombination	0.0069	0.0205	0.3386	0.7349	rosenkopf_top5
recombination_degree	-0.0597	0.2786	-0.2143	0.8303	rosenkopf_top5
n_prior_art_cites	0.0055	0.0041	1.3577	0.1746	rosenkopf_top5
total_patents_same_year	-0.0001	0.0001	-0.9622	0.3360	rosenkopf_top5
total_patents_until_prev_log	-0.4208	0.1786	-2.3566	0.0184	rosenkopf_top5

Full coefficient dump exported from the local estimation script.

K Appendix K: Missingness Audit

Variable	Missing	MissingShare
Inventive impact	0.0000	0.0000
Top-5 indicator	0.0000	0.0000
Breadth of recombination	0.0000	0.0000
Recombination degree	0.0000	0.0000
Field experience	0.0000	0.0000
Organizational experience	0.0000	0.0000
Prior-art citations	0.0000	0.0000
Same-year patents	0.0000	0.0000
Prior cumulative stock	0.0000	0.0000

Missingness audit for the variables summarized in the manuscript.

L Appendix L: Coverage of the Most Observed Entities

Entity	Observations	FirstPeriod	LastPeriod
6066.0000	3200.0000	1986.0000	2016.0000
5047.0000	1318.0000	1986.0000	2016.0000
4194.0000	1017.0000	1986.0000	2010.0000
7435.0000	854.0000	1986.0000	2016.0000
4087.0000	830.0000	1986.0000	2016.0000
4839.0000	815.0000	1986.0000	2016.0000
7585.0000	774.0000	1986.0000	2016.0000
10499.0000	755.0000	1986.0000	2016.0000
5073.0000	671.0000	1986.0000	2008.0000
4060.0000	574.0000	1986.0000	2016.0000
1300.0000	551.0000	1986.0000	2016.0000
8762.0000	524.0000	1986.0000	2016.0000
1690.0000	452.0000	1990.0000	2016.0000
1581.0000	421.0000	1986.0000	2004.0000
10983.0000	403.0000	1986.0000	2016.0000
12141.0000	398.0000	1995.0000	2016.0000
4503.0000	344.0000	1986.0000	2016.0000
6266.0000	340.0000	1986.0000	2016.0000
2285.0000	334.0000	1986.0000	2016.0000
7475.0000	300.0000	1986.0000	1998.0000

Top entities by local observation count.

M Appendix M: Claim-Evidence Audit

Claim	EvidenceType	EvidenceFile	AuditNote
The source paper studies: The source paper asks whether boundary-spanning search improves inventive impact relative to local search.	Direct source fact	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/patent literature/Strategic Management Journal - 2001 - Rosenkopf - Beyond local search boundary-spanning exploration and impact in the.pdf	Recovered from the source article before local writing begins.

Claim	EvidenceType	EvidenceFile	AuditNote
The local executable panel contains 23,844 observations across 184 entities.	Local calculated fact	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_analysis_dataset.csv — patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_sample_summary.csv	Backed by the verified analysis panel and sample audit.
The local headline estimate is -0.0389 for boundary_existing_field vs local with p = 0.3868.	Local calculated fact	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_model_coefficients.csv	Backed by the exported coefficient file used in the paper.
The manuscript uses local substitutions or analogues wherever the original data architecture is not fully recoverable.	Replication decision	Source-vs-decision table and limitations section	Used only when the original pipeline cannot be matched byte-for-byte from current local files.

Expanded claim-evidence audit.

N Appendix N: Build Notes

Build Notes

- batch date: ‘2026-04-04‘ - manuscript style: long-form journal-style replication draft - evidence rule: only processed local datasets and actual local estimates are used - workflow rule: read the source paper first, then map required datasets to verified local files, then write the paper from those files - skill path guiding the structure: ‘C:/Users/bai/.codex/skills/write-replication-journal-github/SKILL.md‘

O Appendix O: Manuscript Metadata

Field	Value
Paper	Rosenkopf and Nerkar (2001)
Source title	Beyond Local Search: Boundary-Spanning, Exploration, and Impact in the Optical Disk Industry
Authors	Lori Rosenkopf and Atul Nerkar
Outlet	Strategic Management Journal
Batch date	2026-04-04
Rows	23,844
Entities	184
Headline term	boundary_existing_field vs local
Headline estimate	-0.0389
Headline p-value	0.3868
Panel path	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_analysis_dataset.csv

Metadata block for GitHub publishing and future reruns.

P Appendix P: Replication Summary Note

Research question: The source paper asks whether boundary-spanning search improves inventive impact relative to local search. Local design: The local analogue reconstructs local, boundary-existing-field, and boundary-new-field search from organizational and field experience counts already materialized in the processed recombination panel. Estimand: Patent-level impact and top-tail models with search-type indicators and patent-level controls. Local result focus: The local analogue supports the most novel boundary-spanning channel much more clearly than the moderate boundary-spanning channel. Comparison note: The local evidence is strongest for boundary-new-field search rather than for boundary-existing-field search. Boundary-new-field patents receive higher average impact relative to local search, while boundary-existing-field patents are lower and imprecise.

References

[1] Lori Rosenkopf and Atul Nerkar. Beyond local search: Boundary-spanning, exploration, and impact in the optical disk industry. *Strategic Management Journal*, 22(4):287–306, 2001.